

HAVANA JOSE MARTI, Cuba

WMO: 782240

Lat: 22.989N

Lon: 82.409W

Elev: 210

StdP: 14.58

Time Zone: -5.00 (CUB)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	50.3	53.7	46.0	46.3	58.1	48.2	50.4	60.1	23.0	71.0	20.9	70.9	1.6	230	0.335

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.2	91.6	76.8	91.1	76.8	89.8	76.7	81.6	86.5	80.3	86.0	79.2	85.5	7.9	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.5	159.7	85.0	79.0	151.5	83.6	77.4	143.6	82.2	45.5	86.0	44.2	86.9	42.9	85.0	93.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.2	18.0	16.1	44.5	93.8	3.0	1.1	42.3	94.6	40.5	95.3	38.8	96.0	36.6	96.8
			43.8	82.0	2.8	3.2	41.8	84.3	40.2	86.2	38.6	88.0	36.6	90.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	76.3	69.3	71.0	72.5	75.9	78.9	80.7	81.4	81.5	80.6	78.3	73.3	71.5	(6)
(7)		DBStd	5.60	5.14	4.87	4.51	3.73	2.48	1.90	1.48	1.45	1.56	3.02	4.21	4.77	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	48	22	11	4	0	0	0	0	0	0	0	1	10	(9)
(10)		CDD50	9589	597	588	698	777	896	922	972	976	919	878	700	666	(10)
(11)		CDD65	4164	154	179	237	327	431	472	507	511	469	413	252	212	(11)
(12)		CDH74	34911	917	1267	1900	2793	3827	4341	4797	4734	4035	3296	1719	1285	(12)
(13)		CDH80	12340	151	290	524	962	1442	1666	2026	2003	1560	1071	374	271	(13)
(14)	Wind	WSAvg	7.0	7.8	7.6	8.5	8.3	7.4	6.3	5.8	5.7	5.6	6.9	7.3	7.1	(14)
(15)	Precipitation	PrecAvg	47.8	1.8	1.5	1.6	3.0	5.1	7.3	5.3	6.2	6.7	5.2	2.2	1.9	(15)
(16)		PrecMax	60.3	6.8	4.8	3.9	5.9	15.7	15.9	8.3	11.3	13.0	14.4	6.0	6.4	(16)
(17)		PrecMin	36.1	0.2	0.3	0.7	0.9	1.5	3.2	2.4	3.7	3.0	1.5	0.1	0.3	(17)
(18)		PrecStd	6.3	1.6	1.2	0.8	1.3	2.9	3.0	1.4	1.6	2.0	2.9	1.5	1.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	85.6	86.4	88.2	90.0	91.8	92.8	92.8	93.0	91.6	90.1	87.9	86.3	(19)
(20)			MCWB	70.6	70.9	71.9	72.4	74.6	77.2	76.8	77.1	77.2	76.7	74.8	73.3	(20)
(21)		2%	DB	83.8	85.6	86.3	89.1	89.9	91.1	91.4	91.5	90.0	88.2	86.0	84.5	(21)
(22)			MCWB	70.8	70.8	72.3	72.6	74.8	77.2	76.8	77.4	77.5	77.0	74.4	72.6	(22)
(23)		5%	DB	82.0	83.9	84.6	87.5	88.7	89.6	89.9	90.0	89.4	87.6	84.1	82.8	(23)
(24)			MCWB	70.6	70.9	71.4	72.5	74.7	77.0	77.0	77.5	77.5	77.0	73.3	71.9	(24)
(25)		10%	DB	79.5	82.0	83.2	85.8	87.4	88.0	89.3	89.4	87.9	86.0	82.2	81.0	(25)
(26)			MCWB	69.6	70.6	70.5	72.1	74.4	77.1	77.1	77.6	77.6	76.5	72.3	71.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	75.1	75.1	77.2	77.4	79.1	81.7	82.1	83.1	82.6	82.2	78.0	76.4	(27)
(28)			MCD B	80.1	81.4	83.8	85.0	86.3	87.3	87.3	88.1	86.5	85.9	82.9	82.1	(28)
(29)		2%	WB	73.5	73.8	75.2	75.9	78.0	80.1	80.3	81.3	80.8	80.0	76.3	74.9	(29)
(30)			MCD B	79.2	80.2	81.8	83.6	85.6	86.2	86.5	86.7	85.7	84.3	82.0	81.0	(30)
(31)		5%	WB	72.3	72.8	73.9	74.8	76.9	79.1	79.1	79.7	79.2	78.6	75.3	73.8	(31)
(32)			MCD B	78.3	79.3	80.8	82.5	85.0	85.4	86.1	86.1	85.1	83.7	80.9	79.7	(32)
(33)		10%	WB	71.0	71.9	72.6	73.8	75.9	78.2	78.2	78.7	78.5	77.4	74.3	72.6	(33)
(34)			MCD B	77.1	78.6	79.6	81.6	83.7	84.7	85.6	85.8	84.7	83.1	79.7	78.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	18.6	20.0	20.4	20.0	17.4	15.8	17.2	17.1	16.0	15.6	17.6	18.1	(35)
(36)			MCDBR	20.2	21.4	21.2	20.9	19.0	17.5	18.1	18.3	17.7	17.2	18.3	19.5	(36)
(37)			MCWBR	9.8	9.8	9.3	8.1	7.4	7.3	7.6	7.8	7.6	7.6	8.6	9.5	(37)
(38)		5% WB	MCD BR	17.7	19.0	18.9	18.6	16.7	15.9	17.2	17.1	15.8	14.9	15.3	16.8	(38)
(39)			MCWBR	9.8	9.3	9.3	8.2	7.5	7.7	7.9	8.4	7.9	7.6	8.2	8.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.359	0.367	0.375	0.400	0.426	0.447	0.455	0.431	0.415	0.404	0.375	0.370	(40)	
(41)		taud	2.524	2.507	2.469	2.402	2.371	2.361	2.329	2.416	2.454	2.467	2.497	2.513	(41)	
(42)		Ebn at Noon	288	292	294	286	277	268	266	274	278	277	280	279	(42)	
(43)		Edh at Noon	30	33	36	38	40	40	41	37	36	34	31	30	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1258	1530	1854	2067	2029	1879	2040	1948	1717	1491	1301	1193	(44)	
(45)		RadStd	98	102	69	70	194	136	94	68	91	107	63	92	(45)	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)	N/A	+1.60	N/A	N/A	N/A	N/A	N/A	N/A	-20	N/A	N/A

Nomenclature: See separate page